

# N+- Network Fundamentals and building network

**[Q] Introduction to Network Devices Theory (1 hour): Overview of switches, routers, hubs, firewalls, and access points. Practical (1 hour): Identify ports and functions of network devices in a setup. Task: Create a table comparing network devices and their features.**

**[Q-1] Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WIFI setup.**

**[ANS]**

| Device            | Main Function                                                                   | Typical Ports                                 |
|-------------------|---------------------------------------------------------------------------------|-----------------------------------------------|
| Switch            | Connects multiple devices within a LAN and forwards data to the correct device. | Ethernet Ports (RJ-45), Console Port          |
| Router            | Connects different networks and provides internet access.                       | WAN Port, LAN Ethernet Ports, Console Port    |
| Hub               | Connects multiple devices and broadcasts data to all connected devices.         | Ethernet Ports (RJ-45)                        |
| Firewall          | Monitors and filters incoming and outgoing network traffic for security.        | WAN Port, LAN Ports, Management Port          |
| Access Point (AP) | Provides wireless connectivity to devices in a network.                         | Ethernet Port, Power Port, Antenna Connectors |

**[Q-2] Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.**

**[ANS]**

### **1. ROUTER**

```
----- |  
ROUTER   |  
|         |  
| [WAN] [LAN1] [LAN2] [LAN3] |  
|         |  
| [Console] [Power] Antennas |  
----- |
```

#### **Ports:**

- WAN Port
- LAN Ethernet Ports
- Console Port
- Power Port
- Antennas

### **2. SWITCH**

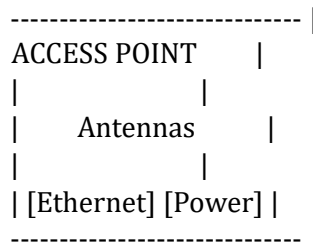
```
----- |  
SWITCH   |  
|         |  
| [Eth1] [Eth2] [Eth3] [Eth4] |  
| [Eth5] [Eth6] [Eth7] [Eth8] |  
|         |  
| [Console] [Power] |  
----- |
```

#### **Ports:**

- Ethernet Ports
- Console Port

- Power Port

### 3. ACCESS POINT



**Ports:**

- Ethernet Port
- Power Port
- Antennas

**[Q-3] Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.**

**[ANS]**

**○ Example: Instagram**

- When I open Instagram on my phone, the **Access Point (WIFI Router)** provides a wireless connection between my phone and the local network.
- The **Router** forwards my internet requests from the local network to Instagram's servers on the internet.
- A **Firewall** helps protect the network by filtering unwanted or malicious traffic while data is being exchanged.
- A **Switch** may be used in larger networks to connect multiple devices and efficiently forward data between them.

**[Q-4] Suppose your college WIFI is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help. Hint: Think about device capacity, coverage, and traffic handling.**

**[ANS]**

- One possible reason for slow and unstable college WIFI is that the **Access Point (AP)** is overloaded with too many connected users. When many students connect to a single AP, bandwidth is shared among all users, causing slow speeds and frequent disconnections.
- Another reason could be limited wireless coverage, resulting in weak signals in some areas of the campus.
- Upgrading to a newer **WIFI 6 Access Point** with higher capacity and better coverage would allow more devices to connect simultaneously and improve network performance.
- This upgrade would reduce congestion, provide faster speeds, and improve connection stability for students.